UNIVERSITY OF VISVESVARAYA COLLEGE OF ENGINEERING
K.R.CIRCLE, BENGALURU – 560001

Department of Computer Science and Engineering

DATABASE MANAGEMENT SYSTEM LABORATORY
MINI PROJECT

Organizational and Personal Budget Management System

Bachelor of Technology

Information Science and Engineering

SUBMITTED BY:

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UNDER THE GUIDANCE OF:

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Introduction

Financial management is an important part of both individual and organizational life. Individuals need to monitor their income, expenses, savings, and monthly budgets, while organizations require a structured system to plan, allocate, monitor, and control financial resources across departments and projects. The Organizational and Personal Budget Management System (OPBMS) is a database-driven application designed primarily for organizational budget management, with personal budget management as an additional feature. The main attraction of the proposed system is its hierarchical organizational budgeting model: Organization → Department → Project → Budget → Expense. This structure allows an organization to allocate its overall budget to departments, allocate departmental budgets to projects, and track expenses against each project.

Problem Statement – Background

Many small organizations and individuals depend on spreadsheets or multiple disconnected applications for managing financial information. These methods can result in data duplication, difficulty in tracking expenses, lack of proper authorization, and difficulty in generating meaningful financial reports. Organizations face additional challenges when budgets are divided between departments and projects. It may be difficult to determine how much money has been allocated, how much has been spent, which projects are exceeding their budgets, and which departments are underutilizing their allocated funds. Therefore, a centralized database management system is proposed to manage organizational budgets, departments, projects, expenses, income, approvals, and reports in an organized manner.

Objective

- Develop a centralized database for organizational and personal financial information.
- Create and manage organizational budgets, and allocate them among departments and projects.
- Record and categorize organizational expenses and income.
- Monitor planned expenditure against actual expenditure, and calculate remaining budgets and utilization.
- Identify departments and projects that exceed their allocated budgets.
- Provide an expense approval mechanism and role-based access to financial information.
- Generate department-wise and project-wise financial reports.
- Provide personal budgeting facilities for individual users, including income, expenses, and savings tracking.

Abstract

In both personal and organizational life, the ability to plan, allocate, and track finances effectively is critical, and a delay or lapse in budget control can lead to overspending and poor financial decisions. To address this, we propose the Organizational and Personal Budget Management System (OPBMS), a database-driven platform that centralizes budgeting and expense tracking across a hierarchical structure of Organization → Department → Project → Budget → Expense. The system allows an organization administrator to create organizations, add departments, define projects under each department, and allocate budgets at every level. Authorized users can record expenses and income against specific projects, while the system automatically calculates amounts spent, remaining budgets, and utilization percentages, and flags budget overruns. An expense approval workflow and role-based access control ensure that financial data remains accurate, authorized, and secure. In addition to the organizational module, OPBMS includes a personal budgeting module that lets individual users manage their own income, expenses, monthly budgets, categories, and savings. By consolidating financial planning, expense tracking, approvals, and reporting into a single system, OPBMS reduces data duplication, improves transparency, and enables organizations and individuals alike to make informed financial decisions.

Technologies used:

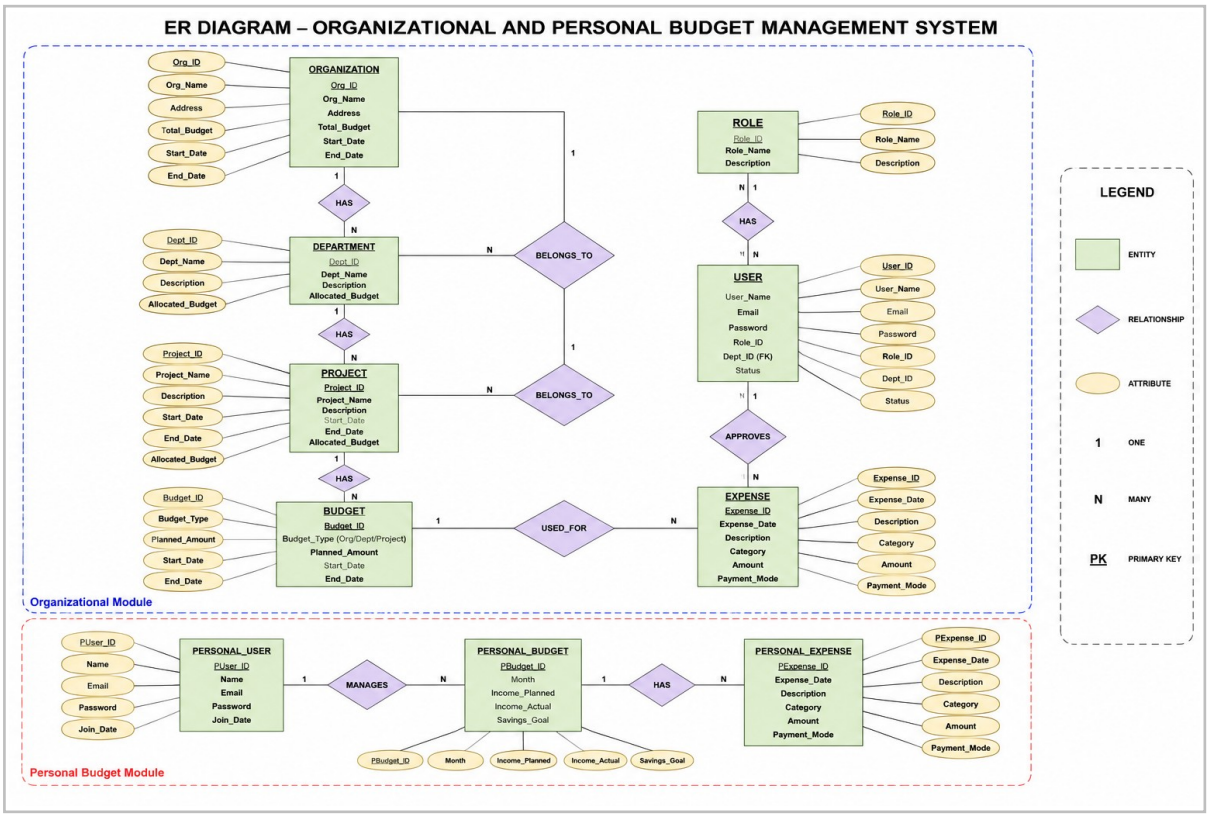
Frontend: HTML, CSS, JavaScript

Backend: Python / Java / PHP, depending on lab requirements

Database: MySQL

Database Design: ER Diagram and Relational Schema

ER Diagram



Notation: Rectangles represent entities, ovals represent attributes, diamonds represent relationships, underlined/marked attributes represent primary keys, FK denotes foreign keys, and 1:N denotes a one-to-many relationship.